

# MTE 544 Assignment 1

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## 1 Question 1: Mobile Manipulator

(a) The transformation matrix between two reference frames is defined as:

$$\mathcal{G}_b^a = \begin{bmatrix} R_b^a & q \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & x_b \\ \sin \theta & \cos \theta & y_b \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where  $\bar{p}^a = \mathcal{G}_b^a \bar{p}^b$  with homogeneous coordinates  $\bar{p} = \begin{bmatrix} p \\ 1 \end{bmatrix}$ .

Starting from the end-effector frame  $\{E\}$  and proceeding back to the global spatial frame  $\{s\}$ :

**1. Transformation from  $\{E\}$  to  $\{2\}$  ( $\mathcal{G}_E^2$ ):** The end-effector frame  $\{E\}$  has its origin displaced by link length  $l_2$  along the  $X_2$  axis. The rotation is fixed at  $\psi = -\frac{\pi}{2}$ :

$$\mathcal{G}_E^2 = \begin{bmatrix} \cos \psi & -\sin \psi & l_2 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\psi = -\frac{\pi}{2}} \mathcal{G}_E^2 = \begin{bmatrix} 0 & 1 & l_2 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

**2. Transformation from  $\{2\}$  to  $\{1\}$  ( $\mathcal{G}_2^1$ ):** Reference frame  $\{2\}$  is displaced by link length  $l_1$  along the  $X_1$  axis with joint rotation  $\gamma$ :

$$\mathcal{G}_2^1 = \begin{bmatrix} \cos \gamma & -\sin \gamma & l_1 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

**3. Transformation from  $\{1\}$  to  $\{B\}$  ( $\mathcal{G}_1^B$ ):** Reference frame  $\{1\}$  is displaced by distance  $l_B$  along the  $X_B$  axis with joint rotation  $\alpha$ :

$$\mathcal{G}_1^B = \begin{bmatrix} \cos \alpha & -\sin \alpha & l_B \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

**4. Transformation from  $\{B\}$  to  $\{s\}$  ( $\mathcal{G}_B^s$ ):** The base frame  $\{B\}$  has its origin at  $(x, y)$  in the global reference frame with heading angle  $\theta$ :

$$\mathcal{G}_B^s = \begin{bmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{bmatrix} \quad (5)$$

**5. Overall Transformation from  $\{E\}$  to  $\{s\}$  ( $\mathcal{G}_E^s$ ):** The overall transformation from frame  $\{E\}$  to the global frame  $\{s\}$  is:

$$\mathcal{G}_E^s = \mathcal{G}_B^s \mathcal{G}_1^B \mathcal{G}_2^1 \mathcal{G}_E^2 \quad (6)$$

Multiplying and simplifying yields:

$$\mathcal{G}_E^s = \begin{bmatrix} \sin(\theta + \alpha + \gamma) & \cos(\theta + \alpha + \gamma) & x + l_B \cos \theta + l_1 \cos(\theta + \alpha) + l_2 \cos(\theta + \alpha + \gamma) \\ -\cos(\theta + \alpha + \gamma) & \sin(\theta + \alpha + \gamma) & y + l_B \sin \theta + l_1 \sin(\theta + \alpha) + l_2 \sin(\theta + \alpha + \gamma) \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

(b) Given the numerical configuration:

$$\theta = -\frac{\pi}{2}, \quad \alpha = \frac{\pi}{6}, \quad \gamma = \frac{\pi}{4}, \quad l_1 = 1, \quad l_2 = 1, \quad l_B = 0.5, \quad x = 0.8, \quad y = 0.8$$

Substituting these values into the overall transformation matrix yields:

$$\mathcal{G}_E^s = \begin{bmatrix} -0.2588 & 0.9659 & 2.2659 \\ -0.9659 & -0.2588 & -0.8248 \\ 0 & 0 & 1.0000 \end{bmatrix} \quad (8)$$

To transform the given points  $p_1^E = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$  and  $p_2^E = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$  to the spatial frame  $\{s\}$ , we express them in homogeneous coordinates  $\bar{p}_i^E = \begin{bmatrix} p_i^E \\ 1 \end{bmatrix}$  and apply  $\bar{p}_i^s = \mathcal{G}_E^s \bar{p}_i^E$ :

$$\bar{p}_1^s = \mathcal{G}_E^s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.2659 \\ -0.8248 \\ 1.0000 \end{bmatrix} \implies p_1^s = \begin{bmatrix} 2.2659 \\ -0.8248 \end{bmatrix} \quad (9)$$

$$\bar{p}_2^s = \mathcal{G}_E^s \begin{bmatrix} 0.5 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.1024 \\ -1.5666 \\ 1.0000 \end{bmatrix} \implies p_2^s = \begin{bmatrix} 3.1024 \\ -1.5666 \end{bmatrix} \quad (10)$$

(c) A target object point is measured in the robot base frame  $\{B\}$  as  $q^B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ , with homogeneous

coordinates  $\bar{q}^B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ .

The transformation from the end-effector frame  $\{E\}$  to the base frame  $\{B\}$  is given by the chain of transformations as we have seen before in part (a):

$$\bar{q}^B = \mathcal{G}_1^B \mathcal{G}_2^1 \mathcal{G}_E^2 \bar{q}^E = \mathcal{G}_E^B \bar{q}^E \quad (11)$$

To express the target point in the end-effector frame  $\{E\}$  for grasping, we invert this transformation. Note this means inverting the matrices as well as the order of multiplication:

$$\bar{q}^E = (\mathcal{G}_E^B)^{-1} \bar{q}^B = (\mathcal{G}_E^2)^{-1} (\mathcal{G}_2^1)^{-1} (\mathcal{G}_1^B)^{-1} \bar{q}^B \quad (12)$$

Evaluating the transformation with the numerical configuration from part (b):

$$\bar{q}^E = (\mathcal{G}_E^2)^{-1}(\mathcal{G}_2^1)^{-1}(\mathcal{G}_1^B)^{-1} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.7418 \\ 0.3542 \\ 1.0000 \end{bmatrix} \quad (13)$$

Thus the location of the point in the end-effector frame  $\{E\}$  is:

$$q^E = \begin{bmatrix} -0.7418 \\ 0.3542 \end{bmatrix} \quad (14)$$

## 2 Question 2: Differential-Drive Simulation

### Simulation Approach

We want to find the pose variables  $x(t), y(t), \theta(t)$ . At each time step, we increment the current position as such:

$$p_{\text{new}} = p_{\text{old}} + \dot{p} \Delta t \quad (15)$$

where,

$$\dot{p} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad (16)$$

- (a) We simulate the differential-drive robot for a duration of 30s with a timestep of  $\Delta t = 0.1$  s. Note it is assumed that the provided velocity is always in the positive x direction in the local frame of the robot. The resulting 2D trajectories ( $x$  vs  $y$ ) and state variables ( $x, y, \theta$  vs time) for each case are shown below:

#### i. Constant Velocities

**Case 1:**  $[v, \omega]^\top = [1, 0]^\top$

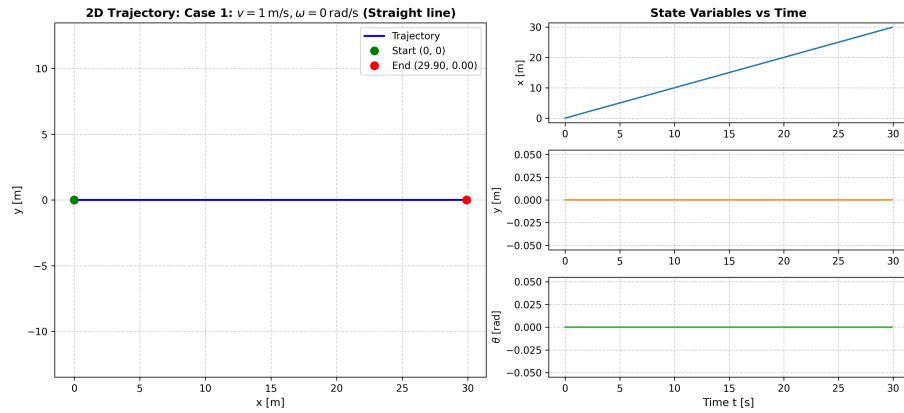


Figure 1: Trajectory and state variables for Case 1

**Observations:** With zero angular velocity ( $\omega = 0$ ), the robot's heading remains constant at  $\theta(t) = 0$  rad. The linear velocity drives the robot in a straight line along the global  $X$ -axis, so we see only  $x$  increase, and  $y$  and  $\theta$  remain constant.

**Case 2:**  $[v, \omega]^\top = [0, 0.5]^\top$

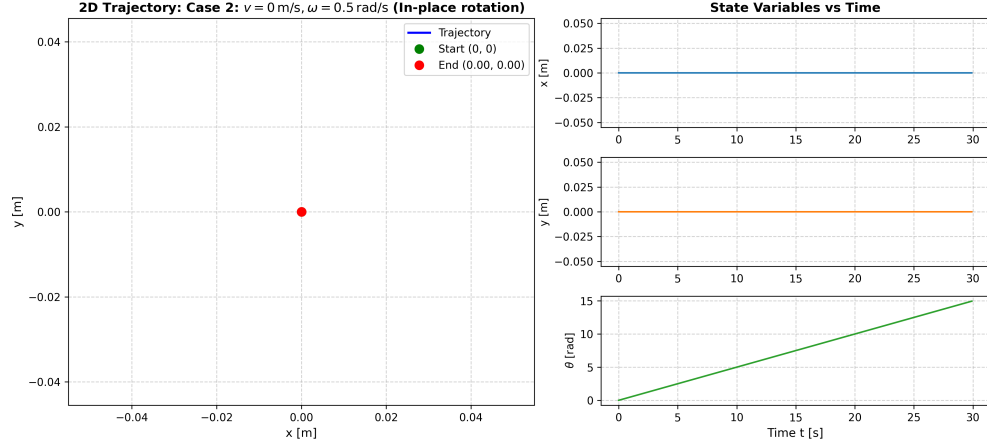


Figure 2: Trajectory and state variables for Case 2

**Observations:** With zero linear velocity ( $v = 0$ ), there is no translation, and the robot rotates strictly in place about its center. The position coordinates don't change. The heading  $\theta(t)$  increases linearly visible only in the  $\theta$  plot.

**Case 3:**  $[v, \omega]^\top = [1, 0.5]^\top$

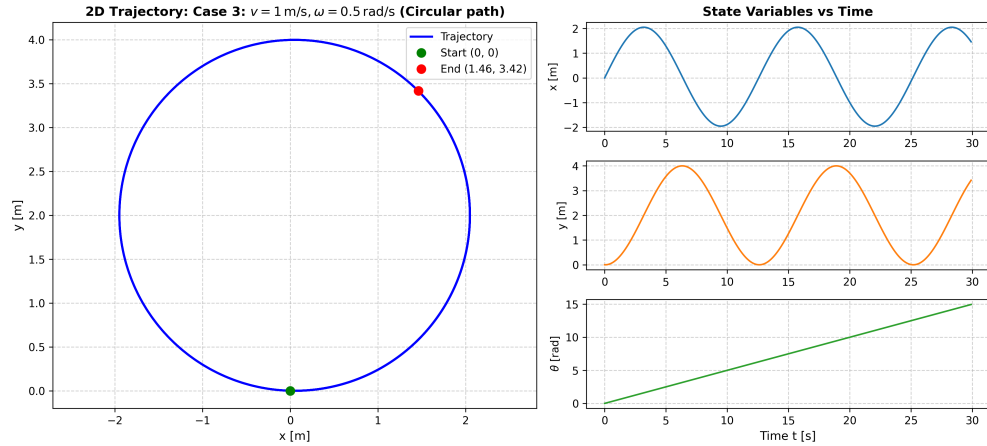


Figure 3: Trajectory and state variables for Case 3

**Observations:** Combining constant linear and angular velocities yields a circle. Particularly the ratio of linear velocity to angular velocity defines the radius of the circle as  $R = \frac{v}{\omega} = \frac{1}{0.5} = 2$  m. We can visually confirm the robot traces a circle of radius 2 m centered at (0, 0) m. The position states  $x(t)$  and  $y(t)$  oscillate sinusoidally with an amplitude of 2 m which is a neat correlation to notice since indeed as we trace around a circle we expect the x and y components to form cos and sin respectively.

## ii. Velocity Profiles

**Profile:** 
$$\begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix} = \begin{bmatrix} 1 + 0.2 \sin(t) \\ 0.2 + 0.6 \cos(t) \end{bmatrix}$$

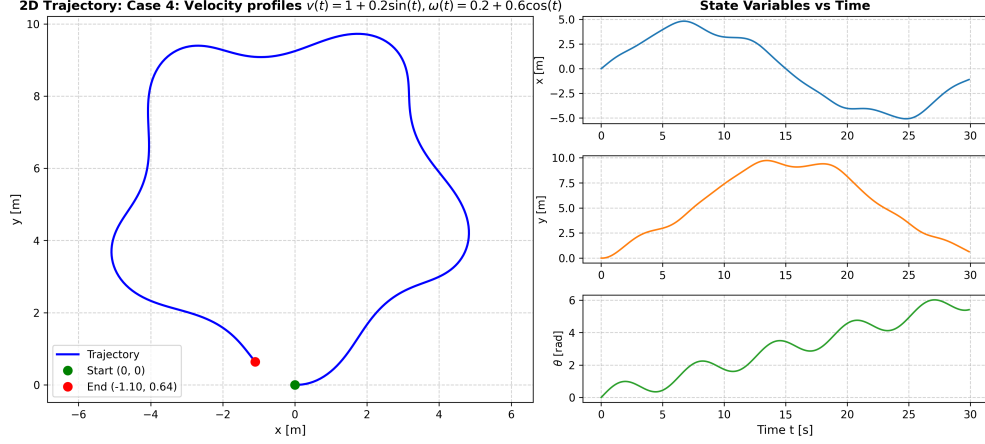


Figure 4: Trajectory and state variables for Case 4.

**Observations:** The angular rate  $\omega(t)$  has a bias/offset of 0.2 rad/s, so it produces an overall net counter-clockwise trajectory while the  $0.6 \cos(t)$  component periodically wiggles the curvature. Coupled with the oscillating speed  $v(t)$ , the robot traces an undulating, complex, yet periodic trajectory.

- (b) Given robot parameters  $T = 0.3$  m (track width) and  $r = 0.15$  m (wheel radius).

The linear velocities at the left and right wheels are:

$$v_L^w = \omega \left( R - \frac{T}{2} \right) = v - \frac{T\omega}{2} \quad (17)$$

$$v_R^w = \omega \left( R + \frac{T}{2} \right) = v + \frac{T\omega}{2} \quad (18)$$

The corresponding wheel speeds  $u_r(v, \omega)$  and  $u_l(v, \omega)$  (in rad/s) are:

$$u_r(v, \omega) = \frac{1}{r} v_R^w = \frac{1}{r} \left( v + \frac{T\omega}{2} \right) = \frac{1}{0.15} (v + 0.15\omega) \quad (19)$$

$$u_l(v, \omega) = \frac{1}{r} v_L^w = \frac{1}{r} \left( v - \frac{T\omega}{2} \right) = \frac{1}{0.15} (v - 0.15\omega) \quad (20)$$

Applying this to the three constant velocity cases:

1. For  $[v, \omega]^\top = [1, 0]^\top$ :

$$u_r(v, \omega) = 6.667 \text{ rad/s}, \quad u_l(v, \omega) = 6.667 \text{ rad/s}$$

2. For  $[v, \omega]^\top = [0, 0.5]^\top$ :

$$u_r(v, \omega) = 0.5 \text{ rad/s}, \quad u_l(v, \omega) = -0.5 \text{ rad/s}$$

3. For  $[v, \omega]^\top = [1, 0.5]^\top$ :

$$u_r(v, \omega) = 7.167 \text{ rad/s}, \quad u_l(v, \omega) = 6.167 \text{ rad/s}$$

**Simulation Modification for Wheel Speed Inputs:** Solving the system of linear equations in (19) and (20):

$$\omega = r \frac{u_r - u_l}{T}, \quad v = r \frac{u_r + u_l}{2} \quad (21)$$

Substituting these directly into the equation from earlier:

$$\begin{aligned} \dot{p} &= \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \\ &= r \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2}(u_r + u_l) \\ \frac{1}{T}(u_r - u_l) \end{bmatrix} \\ &= r \begin{bmatrix} \frac{1}{2} \cos \theta (u_r + u_l) \\ \frac{1}{2} \sin \theta (u_r + u_l) \\ \frac{1}{T}(u_r - u_l) \end{bmatrix} \\ &= r \begin{bmatrix} \frac{1}{2} \cos \theta & \frac{1}{2} \cos \theta \\ \frac{1}{2} \sin \theta & \frac{1}{2} \sin \theta \\ \frac{1}{T} & -\frac{1}{T} \end{bmatrix} \begin{bmatrix} u_r \\ u_l \end{bmatrix} \end{aligned} \quad (22)$$

This matrix expression for  $\dot{p}$  in terms of  $[u_r, u_l]^\top$  now substitutes in exactly where  $\dot{p}$  was computed previously in the Forward Euler update loop ( $p_{\text{new}} = p_{\text{old}} + \dot{p} \Delta t$ ).

### 3 Question 3: 3-Omnidirectional Simulation

- (a) To derive the equations of motion for the 3-wheel omnidirectional robot, we define the coordinate frames and geometric parameters as shown in Figure 5.

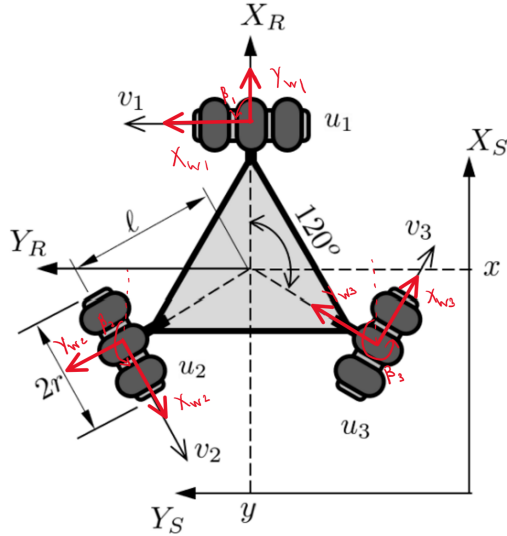


Figure 5: 3-Omnidirectional robot wheel frame assignments.

From the lecture derivation for omnidirectional wheeled mobile bases, the rotational speed  $u_i$  of wheel  $i$  is related to the robot velocity  $\dot{q} = [\dot{x} \ \dot{y} \ \dot{\theta}]^\top$  by:

$$u_i = \frac{1}{r_i} [1 \ \tan \gamma_i] g_i(\theta) \dot{q} \quad (23)$$

where  $g_i(\theta)$  is

$$g_i(\theta) = \begin{bmatrix} \cos(\theta + \beta_i) & \sin(\theta + \beta_i) & X_i \sin \beta_i - Y_i \cos \beta_i \\ -\sin(\theta + \beta_i) & \cos(\theta + \beta_i) & X_i \cos \beta_i + Y_i \sin \beta_i \end{bmatrix} \quad (24)$$

For our symmetrical 3-omniwheel configuration:

- All three wheels share identical radius:  $r = r_1 = r_2 = r_3$ .
- The roller angle for omniwheels is:  $\gamma = \gamma_1 = \gamma_2 = \gamma_3 = 0^\circ$ .
- The wheel orientation angles (measured from the positive x-axis of the robot frame to the positive x-axis of the wheel frame, shown in Figure 5) are separated by  $120^\circ$ :

$$\beta_1 = 90^\circ, \quad \beta_2 = 210^\circ, \quad \beta_3 = 330^\circ$$

Since  $\tan \gamma_i = \tan(0^\circ) = 0$ , only the first row of  $g_i(\theta)$  contributes to driving the wheel:

$$\begin{aligned} u_i &= \frac{1}{r} [1 \ 0] g_i(\theta) \dot{q} \\ &= \frac{1}{r} \left( \dot{x} \cos(\theta + \beta_i) + \dot{y} \sin(\theta + \beta_i) + \dot{\theta} (X_i \sin \beta_i - Y_i \cos \beta_i) \right) \end{aligned} \quad (25)$$

Each wheel is mounted at a distance  $L$  from the robot center. Let  $\alpha_i$  be the angle between the positive x-axis of the robot frame and the line connecting the robot center to the center of the wheel, which is to say the angle to  $Y_i$ . Then  $\alpha_i = \beta_i - 90^\circ$ . The wheel center coordinates  $(X_i, Y_i)$  are:

$$\begin{aligned} X_i &= L \cos \alpha_i = L \cos(\beta_i - 90^\circ) = L \sin \beta_i \\ Y_i &= L \sin \alpha_i = L \sin(\beta_i - 90^\circ) = -L \cos \beta_i \end{aligned}$$

Substituting  $(X_i, Y_i)$  into (25):

$$\begin{aligned} X_i \sin \beta_i - Y_i \cos \beta_i &= (L \sin \beta_i) \sin \beta_i - (-L \cos \beta_i) \cos \beta_i \\ &= L (\sin^2 \beta_i + \cos^2 \beta_i) \\ &= L \end{aligned} \quad (26)$$

This neatly simplifies to the constant distance  $L$  for every wheel, which makes sense intuitively because all wheels are symmetrically located at distance  $L$  from the center.

**Individual Wheel Equations:** Substituting (26) back into (25) gives the general relation:

$$u_i = \frac{1}{r} \left( \dot{x} \cos(\theta + \beta_i) + \dot{y} \sin(\theta + \beta_i) + L \dot{\theta} \right) \quad (27)$$

Plugging in the angles  $\beta_1 = 90^\circ$ ,  $\beta_2 = 210^\circ$ ,  $\beta_3 = 330^\circ$  for each wheel:

$$u_1 = \frac{1}{r} \left( \dot{x} \cos(\theta + 90^\circ) + \dot{y} \sin(\theta + 90^\circ) + L\dot{\theta} \right) \quad (28)$$

$$u_2 = \frac{1}{r} \left( \dot{x} \cos(\theta + 210^\circ) + \dot{y} \sin(\theta + 210^\circ) + L\dot{\theta} \right) \quad (29)$$

$$u_3 = \frac{1}{r} \left( \dot{x} \cos(\theta + 330^\circ) + \dot{y} \sin(\theta + 330^\circ) + L\dot{\theta} \right) \quad (30)$$

Collecting all three equations into matrix form yields:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} G \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \frac{1}{r} \begin{bmatrix} \cos(\theta + 90^\circ) & \sin(\theta + 90^\circ) & L \\ \cos(\theta + 210^\circ) & \sin(\theta + 210^\circ) & L \\ \cos(\theta + 330^\circ) & \sin(\theta + 330^\circ) & L \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \quad (31)$$

(b) To simulate the robot, we need the state derivatives  $\dot{q} = [\dot{x} \ \dot{y} \ \dot{\theta}]^\top$  as functions of the wheel

speeds  $u = [u_1 \ u_2 \ u_3]^\top$ . Inverting the forward kinematic relation  $\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} G \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$  yields:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = r G^{-1} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (32)$$

Inverting  $G$  symbolically in MATLAB gives the inverse kinematic matrix:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \frac{r}{3} \begin{bmatrix} -2 \sin \theta & -2 \cos \left( \theta + \frac{\pi}{6} \right) & 2 \sin \left( \theta + \frac{\pi}{3} \right) \\ 2 \cos \theta & -2 \cos \left( \theta - \frac{\pi}{3} \right) & -2 \cos \left( \theta + \frac{\pi}{3} \right) \\ \frac{1}{L} & \frac{1}{L} & \frac{1}{L} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (33)$$

**I. Rotation Inputs:**  $u_1 = -2.0$  rad/s,  $u_2 = 1.0$  rad/s,  $u_3 = 1.0$  rad/s.

We simulate the motion of the robot for 30s with a timestep of  $\Delta t = 0.1$  s, using  $r = 0.10$  m and  $L = 0.25$  m starting from  $(x, y, \theta) = (0, 0, 0)$ .

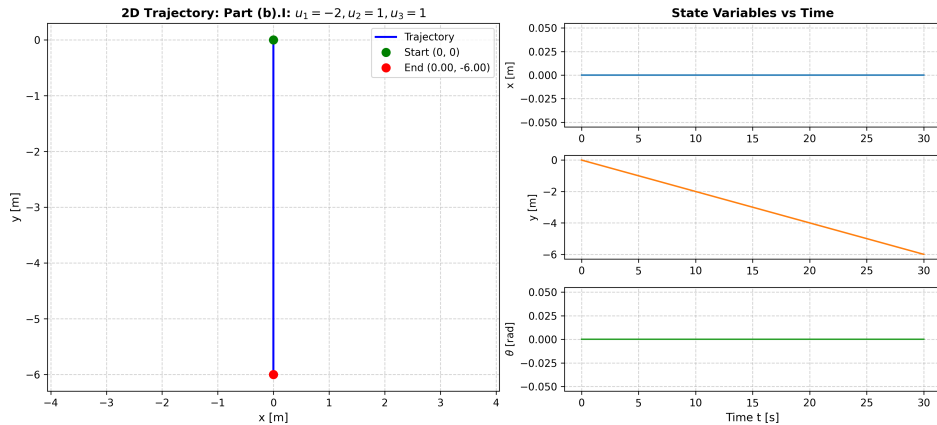


Figure 6: Trajectory and state variables for Part (b).I.

**Observations:** Since the sum of wheel speeds is  $u_1 + u_2 + u_3 = -2 + 1 + 1 = 0$ , the net angular velocity  $\dot{\theta} = \frac{r}{3L}(u_1 + u_2 + u_3) = 0$  rad/s. The robot doesn't rotate at all, so its heading stays flat at  $\theta(t) = 0$  rad. This makes intuitive sense as we can imagine each of the three wheels canceling each other out.

Along the  $y$ -axis, all three wheels push together in the  $-y$  direction to give a constant speed. As expected, the robot traces a straight line along the  $-y$  axis.

## II. Designed control signals

### 1. Straight line with a slope of 60 degrees

To move in a straight line with a slope of  $60^\circ$ , the trajectory must satisfy:

$$\frac{\dot{y}}{\dot{x}} = \tan(60^\circ) = \sqrt{3} \implies \dot{y} = \sqrt{3} \dot{x} \quad (34)$$

Choosing a constant linear velocity along the  $x$ -axis of  $\dot{x} = 1$  m/s yields:

$$\dot{x} = 1 \text{ m/s} \quad (35)$$

$$\dot{y} = \sqrt{3} \text{ m/s} \quad (36)$$

$$\dot{\theta} = 0 \text{ rad/s} \quad (37)$$

Since the robot starts at orientation  $\theta(0) = 0$  and  $\dot{\theta} = 0$ , the orientation remains  $\theta(t) = 0$  for all time. Substituting  $\theta = 0$ ,  $\dot{q} = [1, \sqrt{3}, 0]^\top$ , and wheel radius  $r = 0.10$  m into the forward kinematics relation  $u = \frac{1}{r}G(0)\dot{q}$ :

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} \cos(0 + 90^\circ) & \sin(0 + 90^\circ) & L \\ \cos(0 + 210^\circ) & \sin(0 + 210^\circ) & L \\ \cos(0 + 330^\circ) & \sin(0 + 330^\circ) & L \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \frac{1}{0.10} \begin{bmatrix} 0 & 1 & L \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & L \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & L \end{bmatrix} \begin{bmatrix} 1 \\ \sqrt{3} \\ 0 \end{bmatrix} \quad (38)$$

Evaluating each wheel velocity:

$$u_1 = \frac{1}{0.10}(0 \cdot 1 + 1 \cdot \sqrt{3} + 0) = \frac{\sqrt{3}}{0.10} = 10\sqrt{3} \approx 17.32 \text{ rad/s} \quad (39)$$

$$u_2 = \frac{1}{0.10} \left( -\frac{\sqrt{3}}{2} \cdot 1 - \frac{1}{2} \cdot \sqrt{3} + 0 \right) = \frac{-\sqrt{3}}{0.10} = -10\sqrt{3} \approx -17.32 \text{ rad/s} \quad (40)$$

$$u_3 = \frac{1}{0.10} \left( \frac{\sqrt{3}}{2} \cdot 1 - \frac{1}{2} \cdot \sqrt{3} + 0 \right) = 0 \text{ rad/s} \quad (41)$$

The simulation is run for  $t = 30$  s with initial state  $q(0) = [0, 0, 0]^T$ .

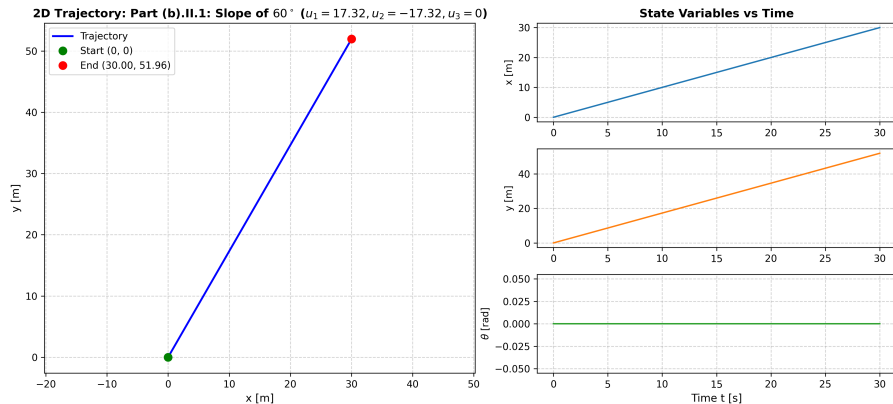


Figure 7: Trajectory and state variables for  $60^\circ$  slope straight line ( $u_1 = 17.32 \text{ rad/s}, u_2 = -17.32 \text{ rad/s}, u_3 = 0 \text{ rad/s}$ ).

**Observations:** With  $u_1 = -u_2$  and  $u_3 = 0$ , the sum of wheel speeds is  $u_1 + u_2 + u_3 = 10\sqrt{3} - 10\sqrt{3} + 0 = 0$ , meaning the net rotational speed is exactly zero and the robot maintains  $\theta(t) = 0$  throughout. Wheel 3 does not spin at all ( $u_3 = 0$ ) and simply rolls freely on its passive rollers as wheels 1 and 2 drive the robot diagonally. Over 30 seconds, the robot reaches  $(x, y) = (30.00, 51.96) \text{ m}$ , maintaining the exact commanded slope of  $51.96/30.00 = 1.732 = \tan(60^\circ)$ .

## 2. Two-metre diameter circle

To trace a circle with a diameter of  $D = 2$  m, the path radius is  $R = 1$  m. As illustrated in Figure 8, we define the instantaneous center of rotation (ICR) such that the robot body frame  $\{R\}$  has its forward axis  $x_R$  tangent to the trajectory with forward linear speed  $v$ , while rotating about the ICR with angular velocity  $\omega = \dot{\theta}$ .

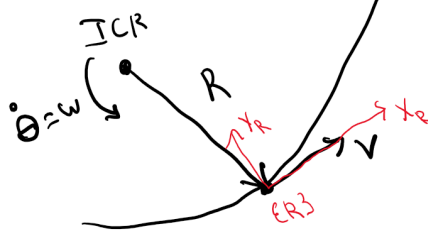


Figure 8: Robot body frame  $\{R\}$  aligned tangentially along the circular trajectory of radius  $R$  about the ICR.

The relationship between linear velocity and angular velocity for circular motion is:

$$v = \omega R \quad (42)$$

With  $R = 1$  m, this simplifies to  $v = \omega$ . Choosing an angular velocity of  $\omega = 1$  rad/s  $= \dot{\theta}$  yields a linear velocity of  $v = 1$  m/s.

In the robot body frame  $\{R\}$ , the velocity components are:

$$\dot{x}^R = v = 1 \text{ m/s} \quad (43)$$

$$\dot{y}^R = 0 \text{ m/s} \quad (44)$$

$$\dot{\theta}^R = \dot{\theta} = 1 \text{ rad/s} \quad (45)$$

To evaluate the forward kinematics, we transform the linear velocity from the robot frame  $\{R\}$  to the global frame  $\{S\}$  via rotation matrix  $R_R^S(\theta)$ :

$$v^S = R_R^S(\theta)v^R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \quad (46)$$

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \quad (47)$$

Substituting  $\dot{q}$  into the forward kinematics relation  $u = \frac{1}{r}G(\theta)\dot{q}$ :

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} \cos(\theta + 90^\circ) & \sin(\theta + 90^\circ) & L \\ \cos(\theta + 210^\circ) & \sin(\theta + 210^\circ) & L \\ \cos(\theta + 330^\circ) & \sin(\theta + 330^\circ) & L \end{bmatrix} \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \quad (48)$$

Simplifying (via MATLAB) the orientation angle  $\theta$  cancels out entirely, yielding constant wheel speeds:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} L \\ L - \frac{\sqrt{3}}{2} \\ L + \frac{\sqrt{3}}{2} \end{bmatrix} \quad (49)$$

Substituting the robot dimensions  $r = 0.10$  m and  $L = 0.25$  m:

$$u_1 = \frac{0.25}{0.10} = 2.5 \text{ rad/s} \quad (50)$$

$$u_2 = \frac{0.25 - \frac{\sqrt{3}}{2}}{0.10} = 2.5 - 5\sqrt{3} \approx -6.16 \text{ rad/s} \quad (51)$$

$$u_3 = \frac{0.25 + \frac{\sqrt{3}}{2}}{0.10} = 2.5 + 5\sqrt{3} \approx 11.16 \text{ rad/s} \quad (52)$$

The simulation is executed for  $t = 30$  s starting from  $q(0) = [0, 0, 0]^T$  with timestep  $\Delta t = 0.1$  s.

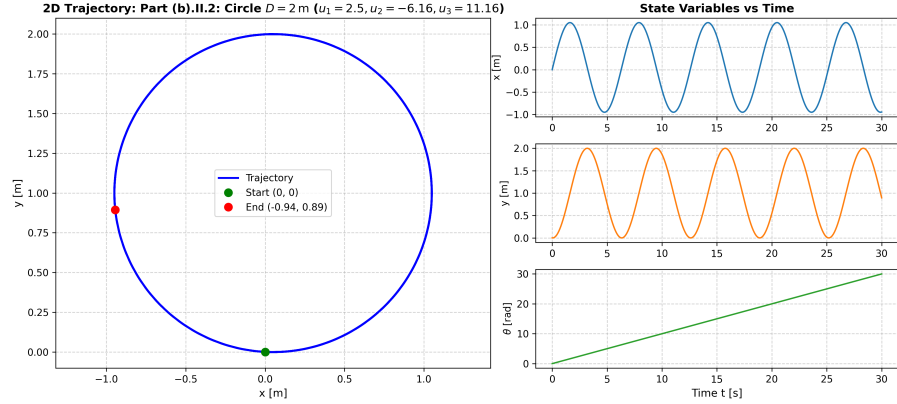


Figure 9: Trajectory and state variables for 2m diameter circle ( $u_1 = 2.5$  rad/s,  $u_2 = -6.16$  rad/s,  $u_3 = 11.16$  rad/s).

**Observations:** Even though the robot traverses a curved circular path, the required wheel inputs  $u_1, u_2, u_3$  are completely constant! This occurs because the robot's heading  $\theta(t)$  rotates at the exact same rate as its tangent velocity vector ( $\dot{\theta} = \omega = 1$  rad/s), keeping the relative velocity of the ground in the robot's local frame invariant. This is more intuitive with a two wheel robot when one wheel is spinning a fixed rate faster than the other, but the same can be accomplished with our 3 omniwheel robot.

Because of the  $\omega / v$  we choose as 1, it completes one full revolution every  $2\pi \approx 6.28$  s. In the Cartesian plane,  $x(t)$  and  $y(t)$  oscillate sinusoidally with amplitude 1.0 m, creating a closed circular path of diameter 2.0 m centered at (0, 1) m. Over the 30-second duration, the robot traces approximately 4.77 ( $\frac{30}{2\pi}$ ) full laps around the circle.